

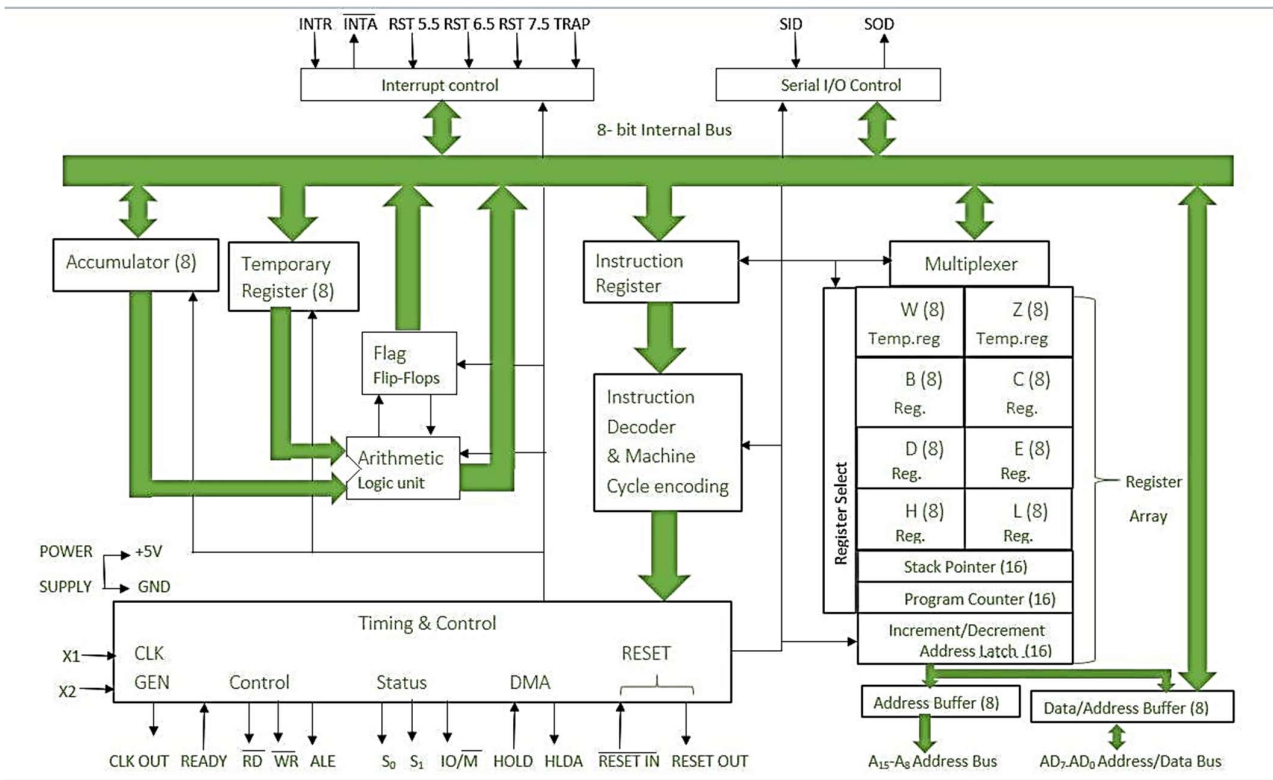
Microprocessor Architecture and Interfacing

Study Guide

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2.1 Microprocessor Architecture and its Operations



The 8085 microprocessor is an 8-bit microprocessor developed by Intel in 1976. It was a popular microprocessor used in embedded systems and early computers.

- **Accumulator (A):**
 - An 8-bit register used in arithmetic, logic, and data transfer operations.
 - Stores the result of ALU operations.
- **ALU (Arithmetic Logic Unit)**
 - Performs arithmetic operations (ADD, SUB) and logical operations (AND, OR, XOR).
 - Inputs come from the accumulator and registers.
- **General Purpose Registers**
 - B, C, D, E, H, L – 8-bit registers.
 - Used to store temporary data during operations.
 - Can be paired as BC, DE, HL to perform 16-bit operations.
- **Flag Register:** Contains 5 condition flags set or reset after an ALU operation:
 - Sign (S) – Set if the result is negative.
 - Zero (Z) – Set if the result is zero.
 - Auxiliary Carry (AC) – Used for BCD operations.
 - Parity (P) – Set if the result has even parity.
 - Carry (CY) – Set if there's a carry out of the MSB.
- **Program Counter (PC)**

- A 16-bit register that holds the address of the next instruction to be executed.
 - Automatically incremented during program execution.
- Stack Pointer (SP)
 - A 16-bit register that points to the top of the stack in memory.
 - The stack is used for temporary storage of data and return addresses during subroutine calls.
- Instruction Register & Decoder
 - Instruction Register holds the current instruction being executed.
 - Decoder interprets the opcode and directs the control unit to execute the instruction.
 - Temporary Register: Used internally by the microprocessor for temporary storage during execution.
- Control Unit:
 - Directs the internal operations of the processor.
 - Generates control signals like RD, WR, IO/M, ALE, etc.
- Timing and Control Unit:
 - Manages all the timing and synchronization of instructions.
 - Coordinates operations between the microprocessor and peripherals.
- Address and Data Buses
 - 16-bit Address Bus: Allows the processor to address 64KB memory.
 - 8-bit Data Bus: Transfers 8 bits of data at a time.
 - Multiplexed Address/Data Bus (AD0–AD7): Lower 8 bits of the address are multiplexed with data for efficient pin use.
- Interrupt System: The 8085 has 5 interrupt inputs:
 - TRAP – Non-maskable, highest priority.
 - RST7.5, RST6.5, RST5.5 – Maskable, vectored.
 - INTR – General-purpose interrupt, non-vectored.
- Serial I/O Control: SID (Serial Input Data) and SOD (Serial Output Data) lines for serial communication using software control.
- Clock System
 - Requires a clock generator (like 8224) if an external crystal is used.
 - Typical frequency: 3 MHz or 6 MHz depending on variant.

Addressing Modes of 8085 Microprocessor

In the 8085 microprocessor, addressing modes refer to the ways in which the operand (data on which the operation is to be performed) is specified in an instruction. These modes define how the microprocessor locates the data required for processing.

Immediate Addressing Mode

- Definition: The operand (data) is given directly in the instruction.
- Syntax: MVI R, data or LXI Rp, 16-bit data
- Example:
 - MVI A, 0x32 → Load the value 32H directly into the accumulator.
 - LXI H, 2500H → Load H-L register pair with address 2500H.

Register Addressing Mode

- Definition: The operand is in a register, and the operation is performed between registers.
- Syntax: MOV R1, R2 or ADD R
- Example:
 - MOV B, C → Copy the content of register C to register B.
 - ADD D → Add content of register D to the accumulator.

Direct Addressing Mode

- Definition: The memory address of the operand is given explicitly in the instruction.
- Syntax: LDA address or STA address
- Example:
 - LDA 2500H → Load the accumulator with the content from memory location 2500H.
 - STA 3000H → Store the content of the accumulator into memory location 3000H.

Indirect Addressing Mode

- Definition: The address of the operand is given indirectly through a register pair (usually HL).
- Syntax: MOV A, M or MOV M, R or LDAX Rp
- Example:
 - MOV A, M → Copy the data from the memory location pointed to by HL into the accumulator.
 - STAX D → Store the content of the accumulator in the memory location pointed to by register pair DE.

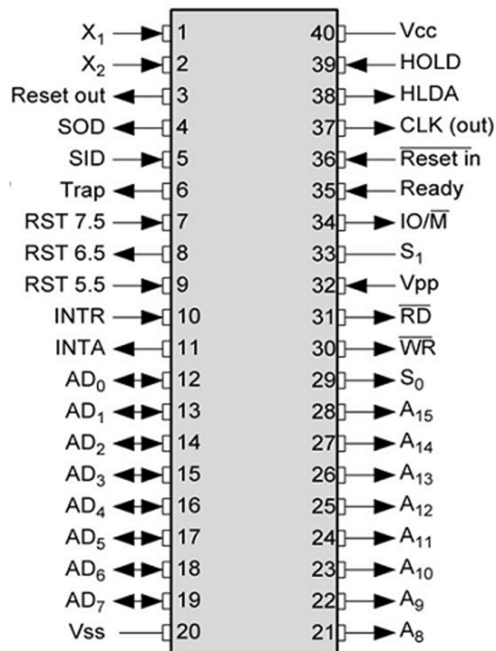
Implied (Implicit) Addressing Mode

- Definition: The operand is implied (not explicitly mentioned) in the instruction. The operation is performed on a specific register or accumulator by default.
- Syntax: e.g., CMA, RAL, INR A
- Example:

- CMA → Complement the contents of the accumulator.
- RAL → Rotate the accumulator left through the carry flag.

Addressing Mode	Example	Description
Immediate	MVI A, 0x32	Operand is part of instruction
Register	MOV B, C	Operand is in register
Direct	LDA 2000H	Address of operand is in instruction
Indirect	MOV A, M	Address is in register pair (e.g., HL)
Implied (Implicit)	CMA, RAL	Operand is implied; usually accumulator or internal register

Pin Diagram of 8085 Microprocessor



The Intel 8085 microprocessor is a popular 8-bit microprocessor. It comes in a 40-pin Dual In-line Package (DIP). Each pin has a specific role for operations like communication, memory access, and control.

Power Supply and Clock Pins:

- Vcc (Pin 40): +5V power supply.
- Vss (Pin 20): Ground (0V reference).
- X1, X2 (Pins 1, 2): Crystal oscillator connections. Typically a 6 MHz crystal is connected to these pins. Internally, the 8085 divides this by 2, resulting in a 3 MHz clock.
- CLK (OUT) (Pin 37): Clock output for synchronizing peripherals.

Address and Data Bus (Multiplexed)

- AD₀–AD₇ (Pins 12–19):

- These are multiplexed lines.
 - During the first part of a machine cycle (T1): They carry the lower-order address (A0–A7).
 - After T1: They carry data (D0–D7).
- A8–A15 (Pins 21–28):
 - High-order address bus.
 - These are dedicated lines for the higher 8 bits of address.

Control and Status Signals

- ALE (Pin 30) – Address Latch Enable:
 - High during T1 to indicate that AD0–AD7 has address information.
 - Used to latch the lower byte of the address.
- RD (Pin 32) – Read:
 - Active low signal.
 - Indicates that the processor is reading from memory or an I/O device.
- WR (Pin 31) – Write:
 - Active low signal.
 - Indicates that the processor is writing to memory or an I/O device.
- IO/M (Pin 34):
 - Distinguishes whether an operation is for memory (low) or I/O device (high).
- S0, S1 (Pins 29, 33):
 - Status signals used for instruction decoding or external devices (not commonly used directly by simple systems).

Interrupts and Externally Initiated Signals

- TRAP (Pin 6):
 - Non-maskable highest priority interrupt.
- RST7.5, RST6.5, RST5.5 (Pins 7, 8, 9):
 - Maskable vectored interrupts.
- INTR (Pin 10):
 - General-purpose interrupt request.
- INTA (Pin 11):
 - Interrupt acknowledge signal (active low).

Other Control Pins:

- RESET IN (Pin 36):
 - Active low input to reset the microprocessor.
- RESET OUT (Pin 3):
 - Indicates the processor is being reset; can reset other devices.
- HOLD (Pin 39):
 - Used to request the microprocessor to release the buses.

- HLDA (Pin 38):
 - Hold Acknowledge — indicates that the processor has released the buses.
- READY (Pin 35):
 - Used to insert wait states. If peripherals are slow, READY can delay the processor.

Pin	Function
1,2	X1, X2 - Clock input
3	RESET OUT
6-9	Interrupt inputs (TRAP, RST7.5, RST6.5, RST5.5)
10	INTR (Interrupt Request)
11	INTA (Interrupt Acknowledge)
12-19	AD0-AD7 (Multiplexed Address/Data)
21-28	A8-A15 (Higher address bits)
29, 33	Status signals S0, S1
30	ALE (Address Latch Enable)
31	WR (Write)
32	RD (Read)
34	IO/M (Memory or I/O select)
35	READY (Peripheral sync)
36	RESET IN
37	CLK (OUT)
38	HLDA (Hold Acknowledge)
39	HOLD (Bus request)
40	Vcc (+5V)
20	Vss (Ground)

2.2 Memory and I/O devices

Memory is used to store instructions and data. In 8085, memory is addressed using 16-bit addresses, allowing access to 64 KB of memory.

Types of Memory

- ROM (Read-Only Memory):
 - Stores firmware or permanent programs like the bootstrap program.
 - Non-volatile (data is retained even after power off).
- RAM (Random Access Memory):
 - Stores temporary data and instructions during execution.
 - Volatile (data is lost when power is off).

Memory Interfacing with 8085

- Address Bus (A0–A15): 16-bit bus for addressing memory locations.
- Data Bus (D0–D7): 8-bit bus to transfer data to/from memory.

Control Signals:

- $\overline{RD}$ (Read): Active low signal, used when data is to be read.
- $\overline{WR}$ (Write): Active low signal, used when data is to be written.
- ALE (Address Latch Enable): Helps separate address and data on AD0–AD7 lines.

Address/Data Bus Multiplexing:

- Lower 8 bits (A0–A7) are multiplexed with D0–D7 to reduce pin count.
- A latch is used to separate the address from the data.

I/O devices are used for communication between the microprocessor and external peripherals like keyboards, displays, printers, etc.

Two Types of I/O Addressing

- Memory-Mapped I/O: I/O devices are treated as memory locations. Uses 16-bit addressing. Allows full memory instructions (like MOV, LDA, STA, etc.). Slower but more flexible.
- I/O-Mapped I/O (Isolated I/O): Separate address space for I/O. Only 8-bit addresses (can address 256 I/O devices). Uses specific I/O instructions like IN and OUT. Faster, but limited instruction set.

I/O Control Signals:

- $IO/\overline{M}$ (I/O or Memory):
 - High → I/O operation.
 - Low → Memory operation.
- $\overline{RD}$ and $\overline{WR}$ are also used with $IO/\overline{M}$ to determine the type of operation:

$IO/\overline{M}$	$\overline{RD}$	$\overline{WR}$	Operation
0	0	1	Memory Read
0	1	0	Memory Write
1	0	1	I/O Read
1	1	0	I/O Write

2.3 Memory interfacing

Memory interfacing in the 8085 microprocessor involves connecting external memory (RAM or ROM) to the microprocessor so that it can read from or write to memory locations during program execution. The 8085 has a 16-bit address bus, so it can access $2^{16} = 65,536$ memory locations (64 KB), each with a unique

address from 0000H to FFFFH.

Address Bus (A0–A15):

- The 8085 has 16 address lines.
- It can directly address 64KB of memory.
- These lines are used to select memory locations.

Data Bus (D0–D7): An 8-bit data bus is used to transfer data to/from memory or I/O.

Control Signals:

- $\overline{RD}$ (Read): Activated (low) when the microprocessor wants to read from memory.
- $\overline{WR}$ (Write): Activated (low) when the microprocessor writes data to memory.
- $IO/\overline{M}$ (I/O or Memory): Distinguishes between I/O and memory operations.
- $IO/\overline{M} = 0 \rightarrow$ Memory operation
- $IO/\overline{M} = 1 \rightarrow$ I/O operation

Memory Chip Selection (Using Address Decoding):

- To connect a memory chip (e.g., 2KB ROM), a decoder or logic gates are used to generate a chip select ($\overline{CS}$) signal.
- This ensures only the desired memory chip is active at a time for a particular address range.

Types of Address Decoding:

- Full Decoding: All 16 address lines are used to generate a unique chip select.
- Partial Decoding: Only some address lines are used; easier but may cause mirror addresses.

Example: Interfacing a 4KB ROM

- ROM size: 4KB = 2^{12} locations $\rightarrow$ needs 12 address lines (A0–A11)
- Remaining address lines (A12–A15) are used for decoding
- If ROM is mapped at address 0000H–0FFFH, then:
 - A15–A12 = 0000
 - Use a logic circuit (like a 4-input NAND gate) to activate $\overline{CS}$ when A15–A12 = 0000

2.4 Interfacing I/O devices

- Input-Output (I/O) Interfacing is a very important concept which acts as a communication medium between the processor, memory unit, and other peripheral devices. I/O interfacing is crucial for ensuring effective transfer of data and information between internal and external components of a computing system.
- I/O interfacing, also referred to as input-output interfacing, is nothing but a way of enabling effective communication between the processor and the peripheral devices like memory, keyboard, mouse, etc.

- I/O interfacing ensures smooth exchange of data and information among different components of the system. It optimizes the system operation by reducing the differences between data transfer speeds, formats, and operations between the processor and other devices.

Functions of I/O Interfaces

I/O interfacing in a microprocessor- or microcontroller-based system performs the following key functions –

- It provides synchronization between the operational speeds of the processing unit and the peripheral devices.
- It allows for selecting an appropriate device for processing input or output signals.
- It also generates control and timing signals.
- It makes use of the data bus to enable data buffering.
- It also identifies errors during exchange of data.
- It also allows for converting serial data into parallel and vice-versa, or digital data into analog signals and vice-versa.

Structure of I/O Interfacing

This structure comprises several components which are explained below in detail:

- **Data Bus Buffer** – This component of the I/O interfacing unit allows bidirectional communication between the processor and the internal system bus. It enables exchange of data, control words, and status information.
- **RD/WR Control Logic** – It is the Read/Write Control Logic block that generates control signals for operation of various system components. It is responsible for directing the flow of data between the processor and the I/O devices. This block determines the direction of data transfer as follows –
 - In read mode, it directs the data flow from input-output devices to the processor.
 - In write mode, it directs the data flow from the processor to the input-output devices.
- **Port A and B** – These ports are interfacing points for connecting input-output devices with the internal system bus. Each of these two ports has a bidirectional buffer for smooth exchange of data.
- **Control and Status Register** – These components hold control signals from the processor and use them to determine the status of the port, data transfer, and errors.

Addressing in I/O Interfacing

The processor uses the address bus for selecting the interfacing unit. For this purpose, two least significant lines of the address bus namely, A0 and A1 are used as the select lines S0 and S1 respectively. These two select lines are used for selecting any of the four interfacing registers namely, port A, port B, control register, or status register.

The selection of a specific interfacing unit is done as per the following criteria

Read State

Chip Select	Operation		Select Lines		Selected I/O Interface
CS	Read (RD)	Write (WR)	S ₀	S ₁	
0	0	1	0	0	Port A
0	0	1	0	1	Port B
0	0	1	1	0	Control Register
0	0	1	1	1	Status Register

Write State

Chip Select	Operation		Select Lines		Selected I/O Interface
CS	Read (RD)	Write (WR)	S ₀	S ₁	
0	1	0	0	0	Port A
0	1	0	0	1	Port B
0	1	0	1	0	Control Register
0	1	0	1	1	Status Register

- For S₀ S₁ = 0 1, the Port B data register is selected for exchange of data between the processor and the I/O devices.
- For S₀ S₁ = 1 0, the control register is selected to store the control information produced by the processor.

Applications of I/O Interfacing:

The following are some key applications of input-output interfacing:

- I/O interfacing allows different peripheral devices to connect to a microprocessor or a microcontroller or a computing system.
- It also allows for efficient file access without any information related to structure of the files.
- I/O interfacing also plays an important role in data acquisition and processing.
- I/O interfacing is used in industrial control system to connect processors with sensors and control

devices.

- I/O interfacing is employed in multimedia systems to handle input and output audio and video signals.
- In network communication systems, I/O interfacing is used to manage data packets, routing, processing, encryption, and decryption.
- I/O interfacing also allows peripheral devices to connect or disconnect to a microprocessor or a microcontroller without affecting the operation of other components.

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